Sadaf Fatima Syeda | SID4 = 2733

SID4=2733, SEED=2733, SLICE=733, HP_ID=3, CLS_A=3, CLS_B=7 (only SEED used)

Task 1 — Prompt Engineering (5 marks)

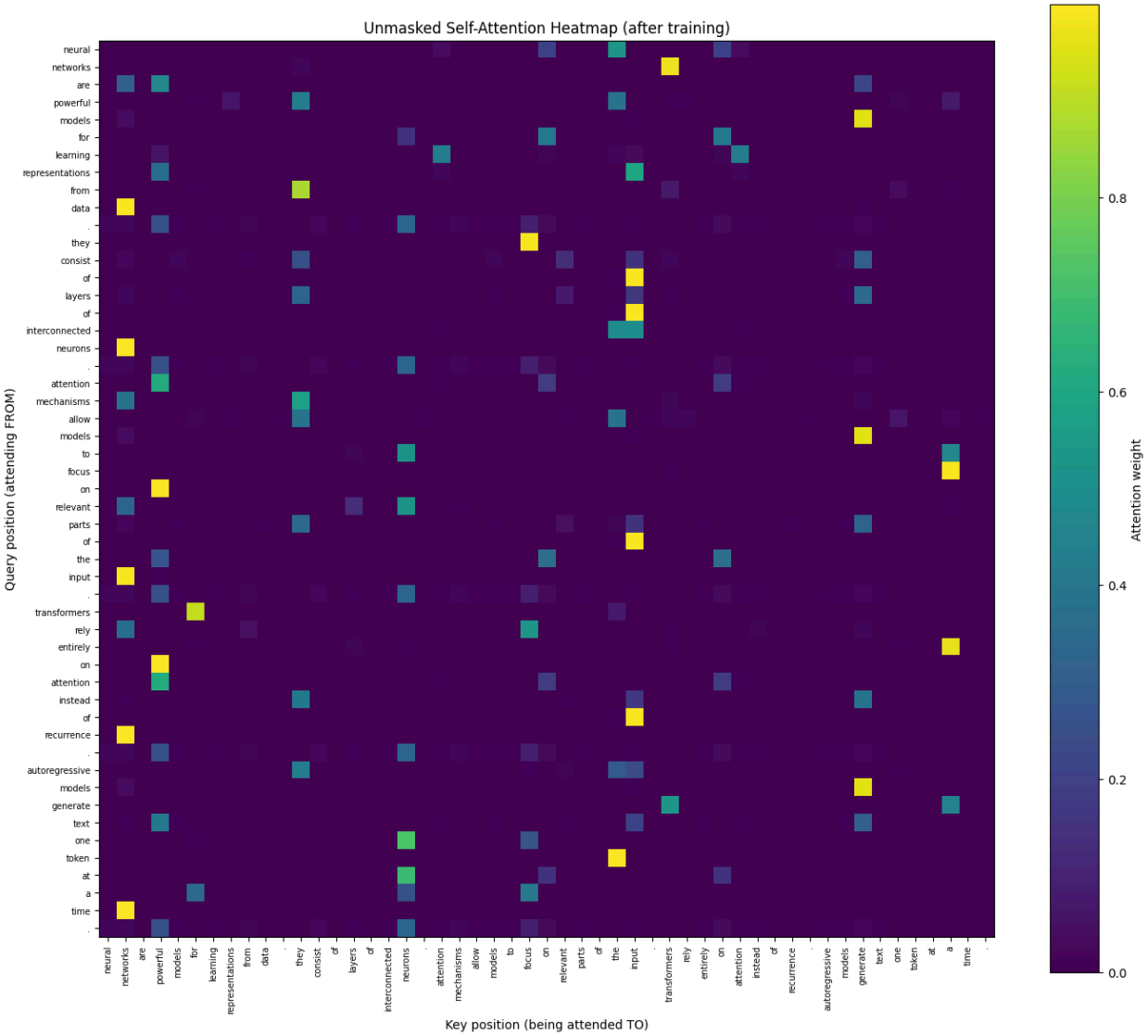
Used Groq API (model=openai/gpt-oss-20b) via LangChain, same 2 questions (1 math, 1 logic) across all 6 techniques.

All six got both questions correct — the questions weren't hard enough to expose correctness differences. Real differences were in reasoning style: Zero-Shot/Few-Shot gave plain prose, CoT and Zero-Shot CoT gave explicit numbered steps (Zero-Shot CoT even used symbolic logic notation on its own), Meta-Prompting named its strategy before solving, and Tree of Thoughts was the only one to generate 3 independent solutions and compare them before answering.

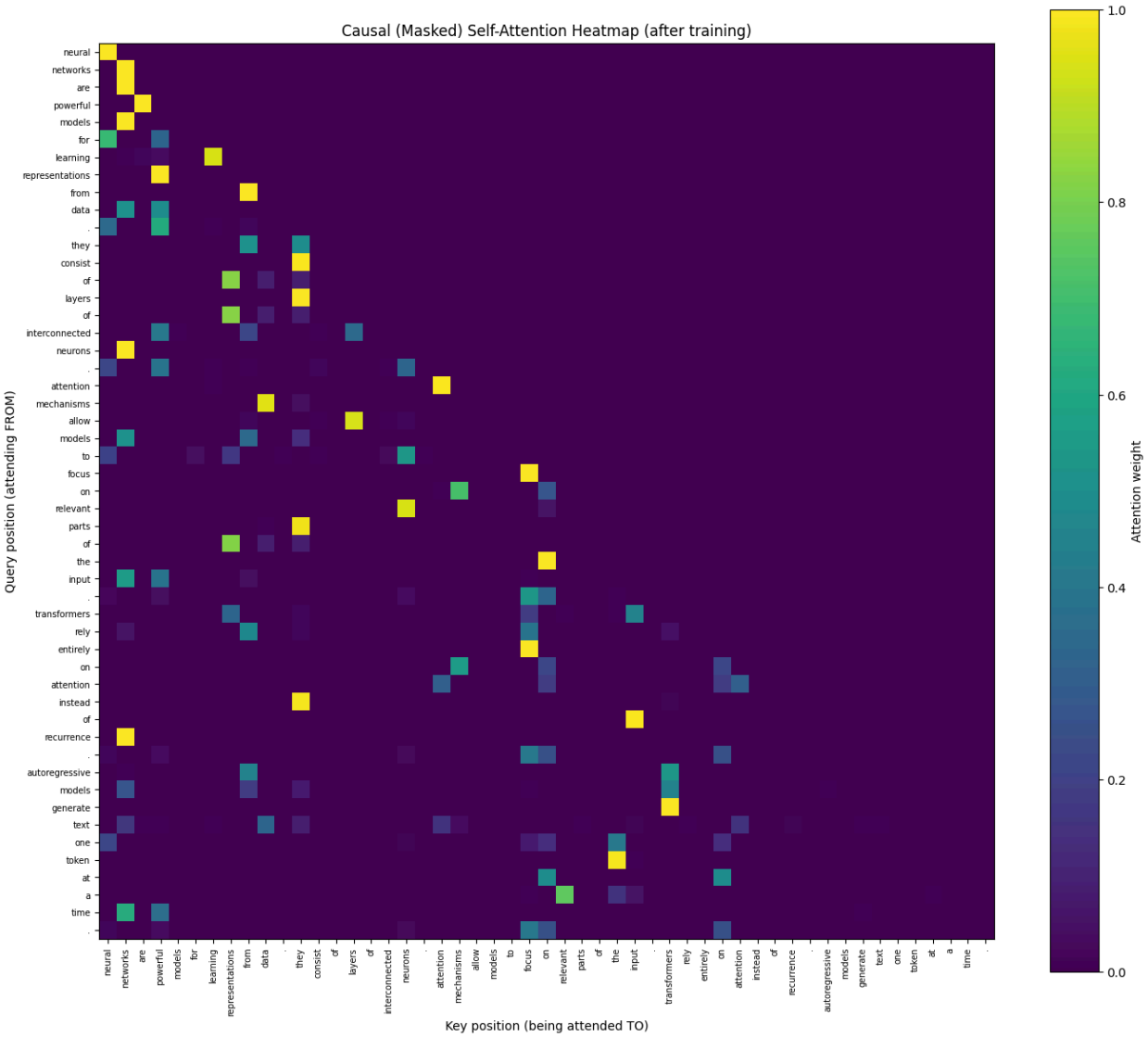
Task 2 — Self-Attention & Causal Masking (10 marks)

Built single-head scaled dot-product attention on the given paragraph (51 tokens, vocab 40). Trained two separate models with next-token prediction:

- **Unmasked:** loss 2.04 to 0.35. Heatmap shows sharp learned peaks (not random), though some tokens over-focus on a few "hub" words — likely overfitting on such a small dataset.



- **Masked:** loss 2.08 to 0.23 (lower than unmasked). Heatmap's upper-right triangle is completely empty — proof no token attends to future position


Causal (Masked) Self-Attention Heatmap (after training)
Query position (attending FROM)
Key position (being attended TO)
Attention weight
neural
networks
are
powerful
models
for
learning
representations
from
data
.
they
consist
of
layers
of
interconnected
neurons
.
attention
mechanisms
allow
models
to
focus
on
relevant
parts
of
the
input
.
transformers
rely
entirely
on
attention
instead
of
recurrence
.
autoregressive
models
generate
text
one
token
at
a
time
.
0.0
0.2
0.4
0.6
0.8
1.0